



# MEDORA

A Smart Healthcare Platform

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# Healthcare Coordination Is Still Manual

## The Fragmented Patient Journey

In Egypt and much of the region, getting care means stitching together disconnected manual steps, each handled by a different actor, with no shared record passing between them.



### Slow, Manual Booking

Booking means finding a clinic phone number and calling during working hours. No standard digital channel exists.



### Lost Paper Prescriptions

Paper is the patient's only treatment record — easily lost, water-damaged, or misread at the pharmacy.



### No Medicine Visibility

Patients cannot tell whether a pharmacy stocks their medicine without physically travelling there, one by one.



# Information & Providers Left Behind



## Unreliable Health Information

Patients turn to general search engines and social media for symptoms, sources with no accuracy filter, where a forum post ranks beside clinical guidance. The result is delayed treatment or inappropriate self-medication.



## Providers Are Digitally Isolated

Small clinics and independent pharmacies, the backbone of primary care, still run on paper notebooks and handwritten ledgers, lacking the budget and tools to offer the digital experience patients now expect.



# Why Build Medora Now



## A Solved Pattern Elsewhere

Technology already fixed this exact coordination problem in banking, food delivery and transport. Healthcare simply has not caught up yet.



## Real, Measurable Harm

Burdensome booking delays care, lost prescriptions break adherence, and bad information drives poor health decisions for chronic patients.



## An Underserved Language

Arabic-first patients are poorly served by tools that bolt on translation as an afterthought, breaking layout and trust.



## Safe AI Is Now Feasible

Modern LLMs can deliver reliable health information, but only inside a carefully engineered safety boundary built for medicine.



## WHAT IS MEDORA?

# One Connected Healthcare Platform

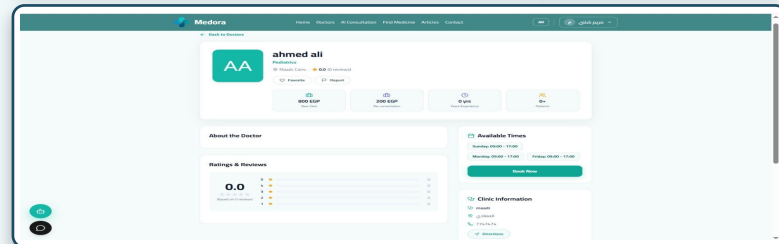
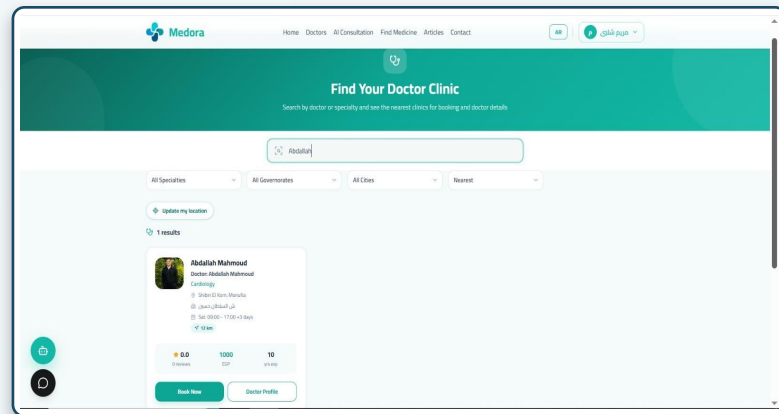
**Medora is not another single-purpose booking app.**

It is a complete, multi-role ecosystem where every participant in the outpatient journey works inside one shared data model, so a booking flows into a digital prescription, and a prescription flows into a real pharmacy



### The promise, in one line

*“Find a verified doctor, book a real appointment, receive your prescription digitally, and order your medicine from a real pharmacy — all in one app, in Arabic and English.”*



# Core Capabilities



## End-to-End Care Flow

Discovery → booking → consultation → digital prescription → medicine order → delivery, all linked in one system.



## Safe Medical AI Assistant

Conversational health information delivered through a five-layer safety pipeline that never diagnoses or prescribes.



## Five Connected Roles

Guest, Patient, Doctor, Pharmacy and Admin — each with a purpose-built interface and strict access boundaries.



## Bilingual by Default

Full Arabic and English with proper right-to-left layout, designed in from the foundation, not bolted on.



# A Three-Tier Distributed System

Each tier has one clear responsibility. The frontend never touches the database, the AI service never changes business data, and the backend never renders UI, clean separation across the whole platform.



## Frontend

React 18 SPA

Presents information and captures user input. Web browser + native Android via Capacitor.



## Backend API

ASP.NET Core .NET 10

All business logic, validation and data access. Controllers → Workflows → EF Core data layer.



## AI Service

Python · FastAPI

Independent microservice for safe medical chat, classification and medicine OCR. Read-only on data.

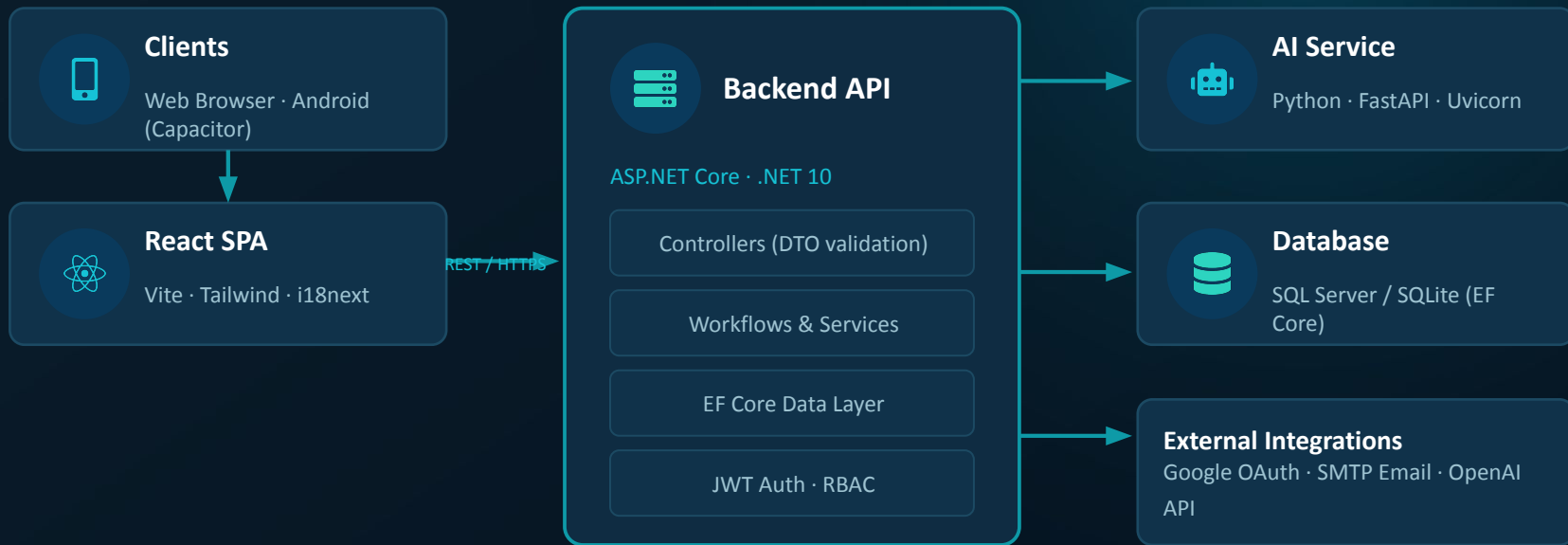


**Data Layer** SQL Server in production · SQLite in development · schema managed entirely through EF Core migrations



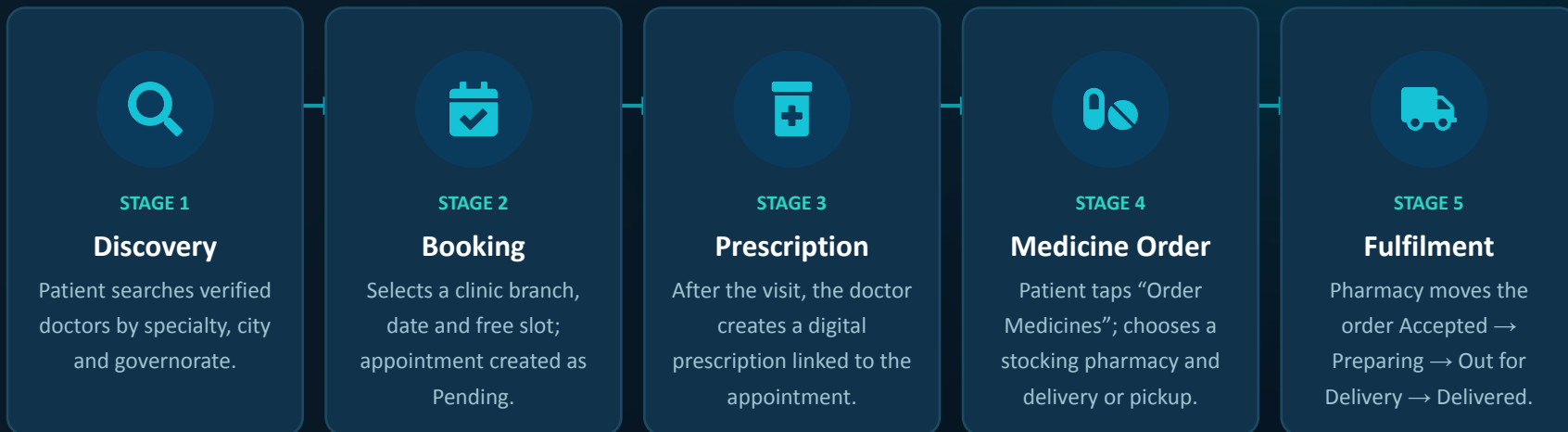
# Component & Deployment Architecture

How the three services, the database, and external integrations communicate



# The Connected Five-Stage Care Flow

Every action creates a permanent, linked digital record, the shared data model that makes true end-to-end coordination possible.



# A Five-Layer Safety Pipeline

Every message is processed through five independent safety layers before any response is generated — off-topic and emergency cases never reach the language model unchecked.



## 1 • Classification

Sorts each message: medical, platform, or non-medical , using keywords plus model reasoning.



## 2 • Off-Topic Filtering

Non-medical messages are rejected instantly and never sent to the API, safer and cheaper.



## 3 • RAG Retrieval

Grounds answers in curated medical CSVs (diseases, drugs, symptoms) to reduce hallucination.



## 4 • Safety Checking

Scans for red-flag symptoms; emergencies trigger an urgent 'seek immediate care' response.



## 5 • Constrained Generation

A strict system prompt forbids diagnosing, dosing, or claiming to give medical advice.

*Owner-checked conversation history keeps every chat private to its creator.*



# Medora Technology Stack

Each service uses the platform best suited to its job



## Frontend

- React 18 + Vite
- TailwindCSS
- react-router-dom
- i18next (AR / EN)
- Leaflet maps
- Capacitor (Android)



## Backend

- ASP.NET Core · .NET 10
- C# 13
- Entity Framework Core
- JWT + ASP.NET Identity
- SQL Server
- Swagger / OpenAPI



## AI Service

- Python 3.11
- FastAPI + Uvicorn
- Pydantic validation
- pandas (medical CSVs)
- OpenAI API (LLM + vision)
- RAG keyword retrieval



# How the Modules Connect

Modules are not isolated — data flows between them automatically to create the integrated experience that is Medora's core value.



A prescription is always linked to a completed appointment.



Prescription items auto-populate the patient's medicine order.



Marking 'Delivered' decrements the pharmacy's stock automatically.



Only a completed appointment unlocks the right to review a doctor.



# Five Roles, One Trust Hierarchy

Medora uses role-based access control. Each role has a defined set of permissions, and the backend enforces data scoping on every query.



## Guest

Browses doctors, medicines and articles freely, no account, no personal actions.



## Patient

Books appointments, views digital prescriptions, orders medicines, writes reviews.



## Doctor

Verified professional, manages clinics, schedule, patients and prescriptions.



## Pharmacy

Verified business, manages inventory and fulfils orders through the delivery workflow.



## Admin

Platform manager, verifies providers, moderates content, resolves reports, audits actions.



# Nine Functional Modules



## Accounts & Auth

Registration, email OTP, login, Google sign-in, password reset, sessions.



## Discovery & Booking

Doctor search with filters, profiles, real-time availability, booking.



## Appointments

Pending → Confirmed → Completed / Cancelled lifecycle for both sides.



## Prescriptions

Doctors create digital prescriptions; patients view, print, share, order.



## Medicine & Pharmacy

Catalogue, detail pages, pharmacy directory and the ordering flow.



## Doctor Dashboard

Clinics, schedule, patients, prescriptions and review management.



## Pharmacy Dashboard

Inventory, order fulfilment, delivery tracking and reporting.



## Admin Panel

Verification, user and content moderation, reports and audit log.

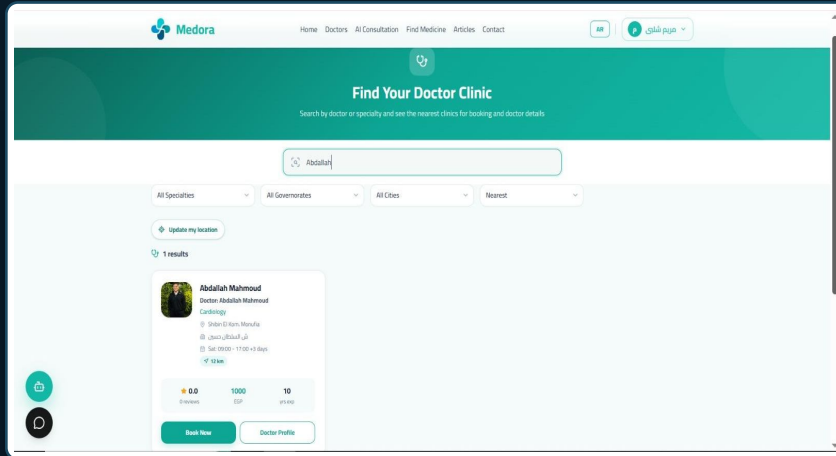


## Medical AI Assistant

Safe conversational health info via the five-layer safety pipeline.

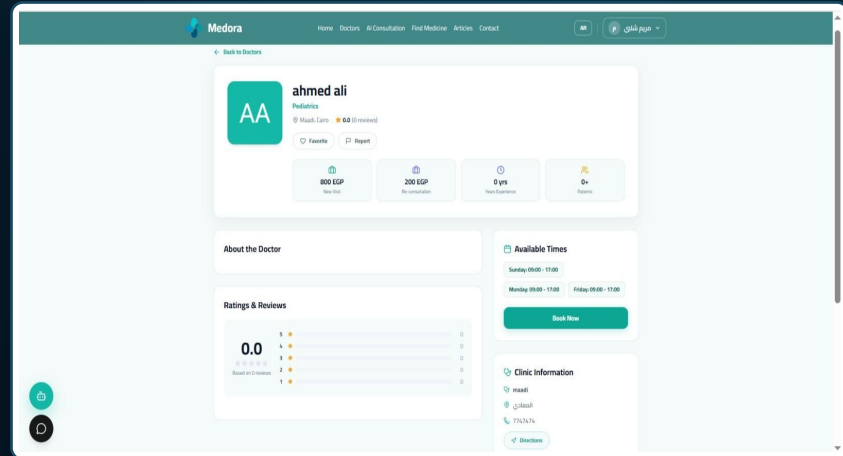


# Patient Experience : Find & Book



## Find a Doctor

Search verified doctors by specialty, city and governorate, with live filters and ratings.

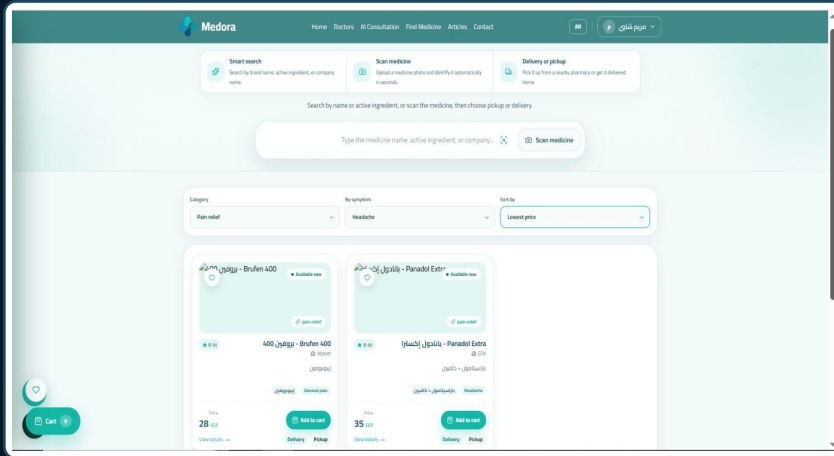


## Doctor Profile & Booking

Real-time slot availability computed from clinic hours and existing bookings, no double-booking.

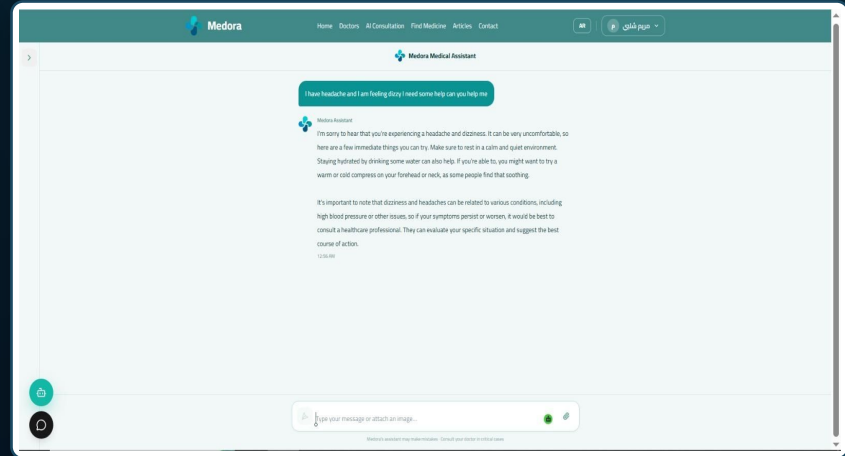


# Medicine Ordering & AI Consultation



## Medicine Search & Order

Search medicines by name or active ingredient, then choose a pharmacy with delivery or pickup.

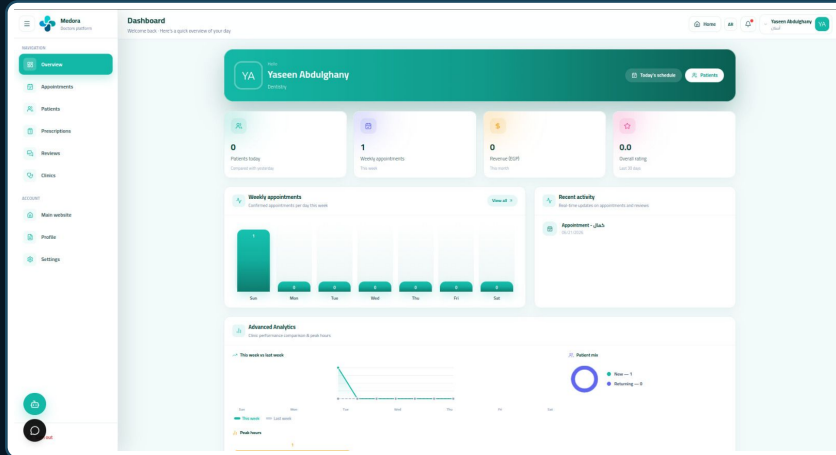


## AI Medical Consultation

A conversational assistant that informs without diagnosing, grounded in curated medical knowledge.

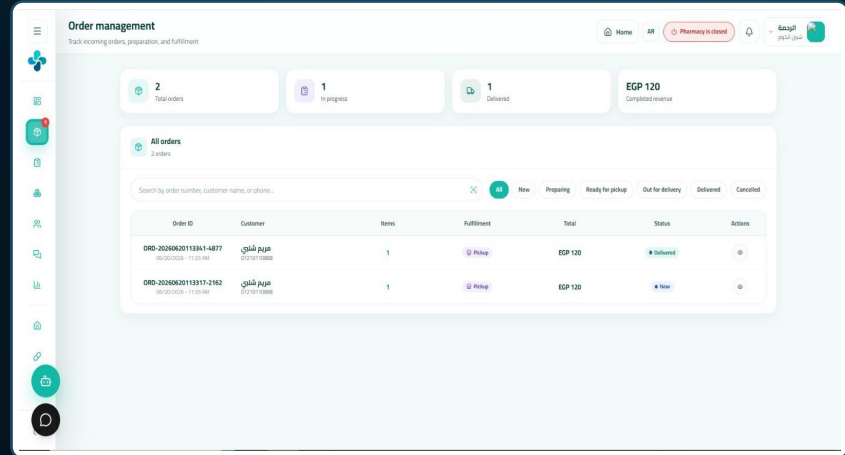


# Provider Dashboards



**Doctor Dashboard**

Overview, clinics, weekly schedule, patient list, prescription creation and review management.



**Pharmacy Order Management**

Incoming orders move through a strict delivery workflow with inventory decremented on completion.



# Defence in Depth

In healthcare the cost of a mistake can be a human life. Security operates at multiple independent layers, each a separate line of defence.



## Transport & CORS

All traffic over HTTPS; CORS restricts the API to registered frontend origins only.



## JWT Auth + Sessions

Signed tokens with configurable expiry; the UserSession table enables 'log out everywhere'.



## Role-Based Authorisation

Every protected endpoint checks the role claim — a patient token cannot reach admin routes.



## Password & OTP Security

PBKDF2/HMAC-SHA256 hashing; OTPs expire and lock after a capped number of wrong attempts.



## Provider Verification

Doctors and pharmacies are activated only after an admin approves their credentials.



## AI Rate Limiting

Guest sessions on the AI service are throttled to prevent abuse of the OpenAI API.



# How Medora Was Validated

## Multi-Level Testing Strategy



### Unit Tests (xUnit)

Medora.Tests verifies workflow and eligibility rules in isolation with in-memory data.



### Integration Tests

Full stack from HTTP request through controller, workflow and EF Core InMemory provider.



### AI Smoke Tests

smoke\_test.py exercises every AI endpoint to confirm the pipeline produces expected output.



### Manual End-to-End

Every patient, doctor, pharmacy and admin journey run in full by the team in their roles.

## Edge Cases Verified

- ✓ Simultaneous slot booking, double-booking prevented
- ✓ Skipping a workflow state, invalid transition rejected
- ✓ Repeated wrong OTP, capped, new code requested safely
- ✓ Weak password, validation rejects with clear feedback
- ✓ Review without completed visit, eligibility blocks it
- ✓ Off-topic AI message, rejected, no model call made



# Achievements & Engineering Challenges

## Key Achievements



### End-to-End Flow Delivered

Discovery to delivery works as one linked system.



### Genuinely Safe Medical AI

Five-layer pipeline blocks diagnosis, dosing and off-topic.



### True Bilingual RTL

Arabic and English as equal first-class languages.



### Verified-Provider Trust

Document verification and review eligibility enforced.

## Hard Problems Solved

### ▶ Single sign-on across 3 services

Shared JWT signing secret, the AI service validates tokens locally.

### ▶ Workflow-violation prevention

Centralised state machines own every status transition.

### ▶ Real-time slots, no double-booking

Availability computed live; bookings enforced atomically.

### ▶ Full right-to-left Arabic layout

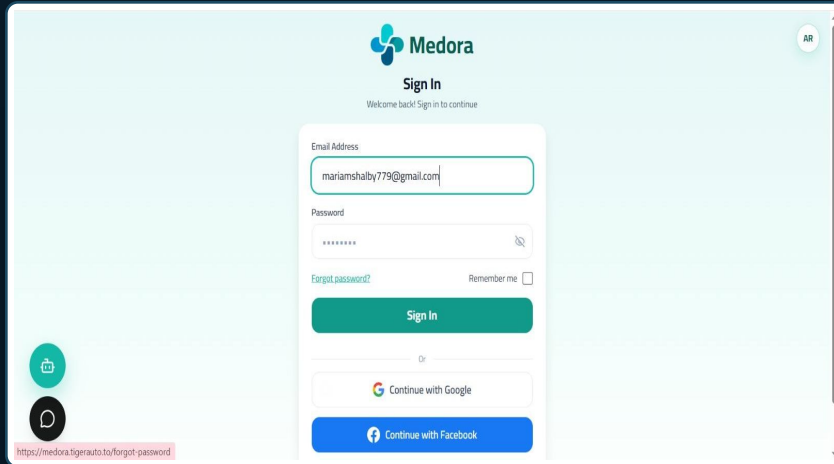
Tailwind logical properties flip with document direction.

### ▶ Fake providers & fake reviews

Admin verification + completed-visit review eligibility.

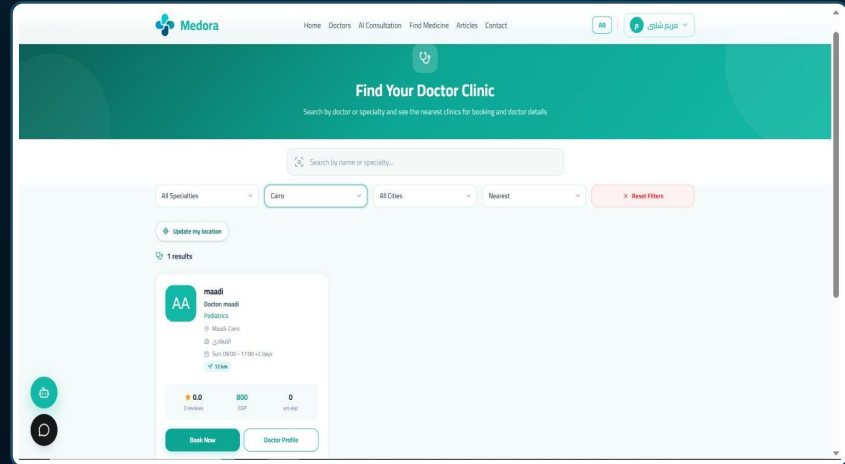


# Patient Access & Discovery



## Authentication

Email / password and Google sign-in, with OTP email verification.

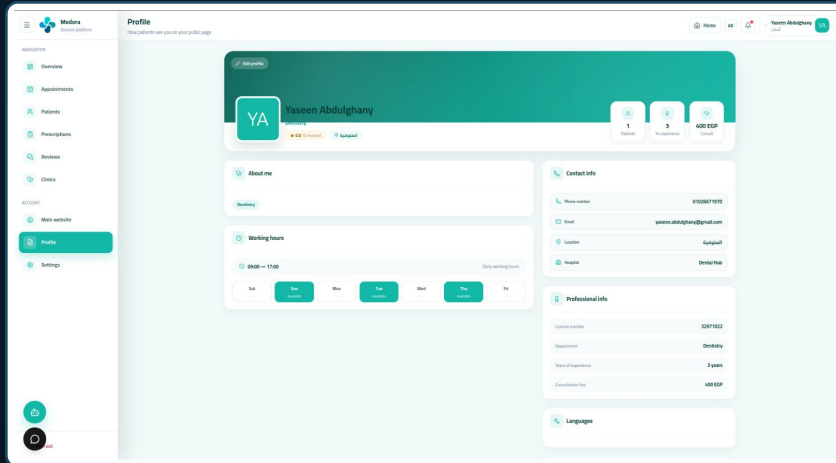


## Doctor Search

Find verified doctors by specialty, city and governorate with live filters.

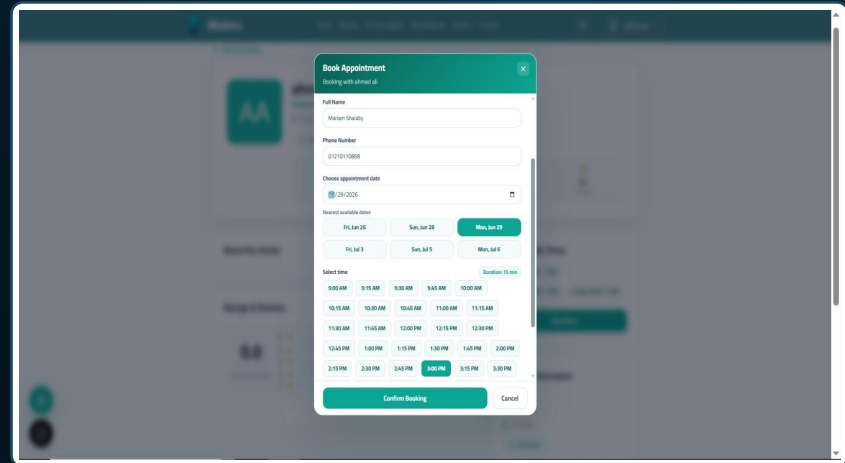


# Doctor Profile & Booking



Doctor Profile

Full profile with fees, ratings, clinic details and available times.

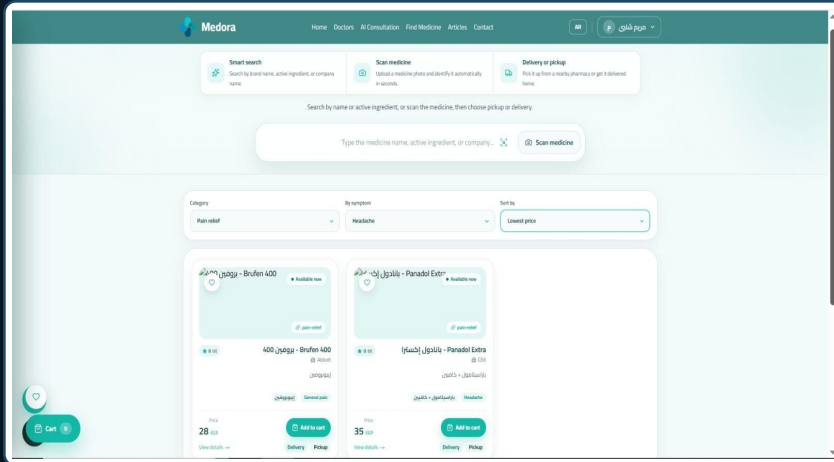


Appointment Booking

Live slot selection that prevents double-booking in real time.

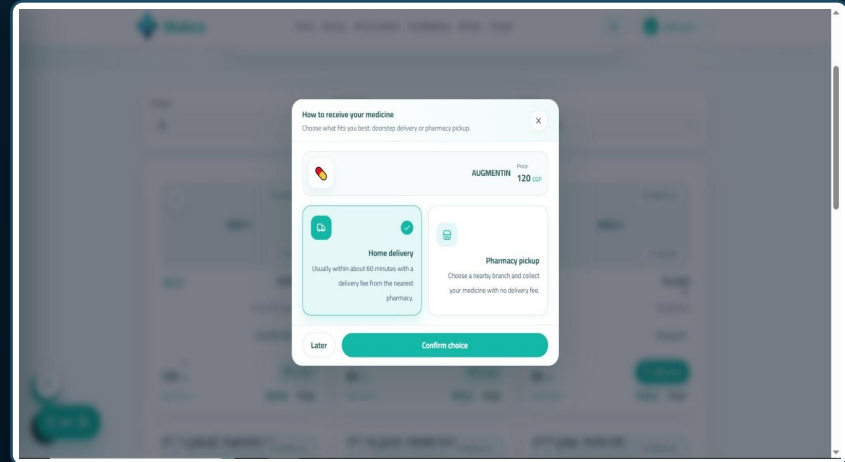


# Medicine Ordering & AI



## Medicine Search

Search medicines by name or active ingredient with pharmacy availability.



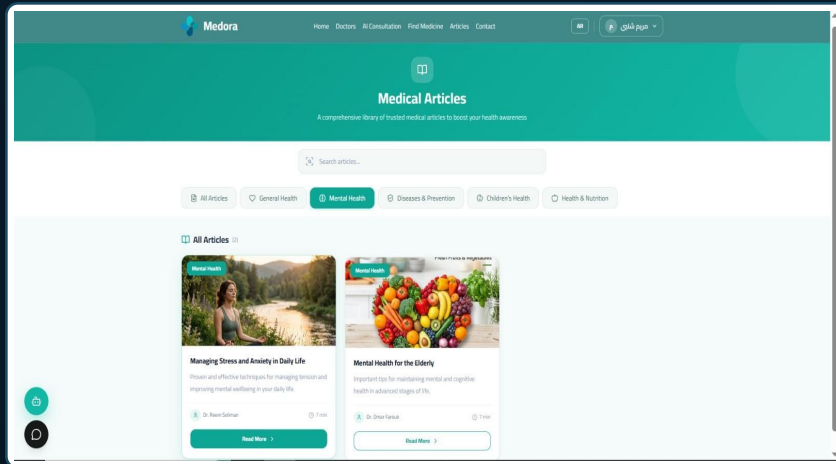
## Delivery or Pickup

Choose home delivery or pharmacy pickup at order checkout.



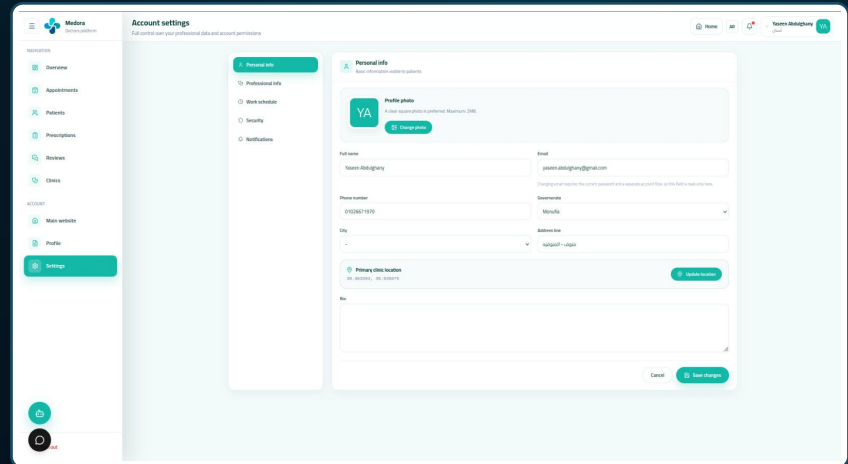
## PROJECT DELIVERABLES

# Health Content & Account



Medical Articles

Verified health-education library, moderated before publication.

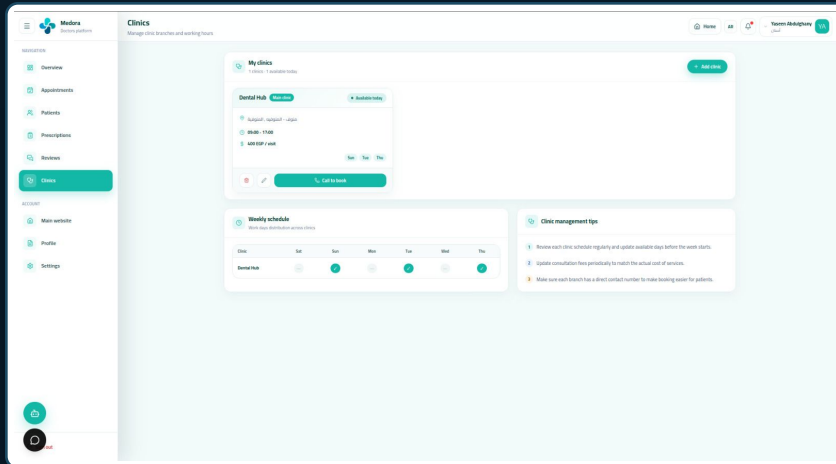


Account Settings

Profile management, language toggle and security controls.

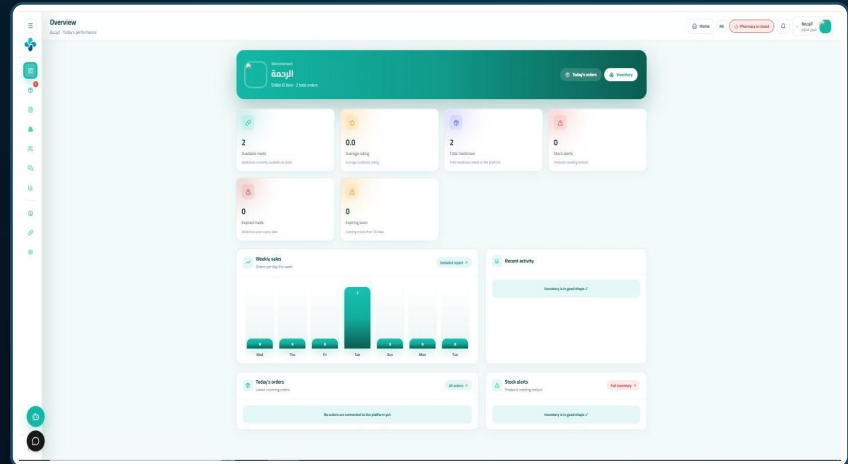


# Provider Operations



Clinics & Schedule

Doctors manage clinic branches and weekly working hours.

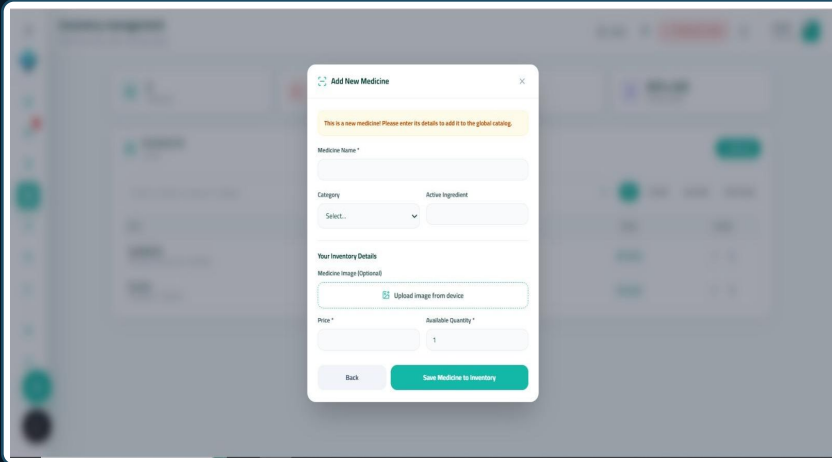


Pharmacy Overview

Order volume, revenue, inventory status and Open/Closed toggle.



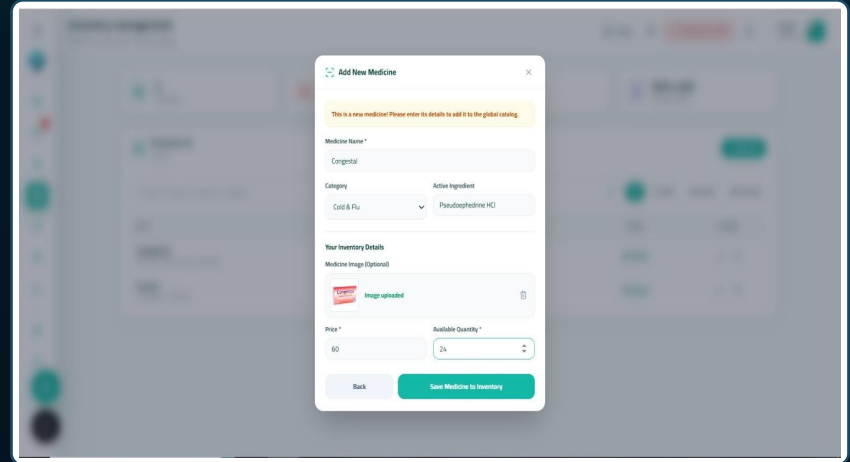
# Admin Catalogue Management



The screenshot shows a mobile application interface with a modal titled "Add New Medicine". At the top, a yellow banner reads "This is a new medicine! Please enter its details to add it to the global catalog." Below this, the form has two main sections. The first section contains a text input for "Medicine Name \*" and a dropdown menu for "Category" with the text "Select..." and a small downward arrow. To the right of the category dropdown is a text input for "Active Ingredient". The second section, titled "Your Inventory Details", contains a text input for "Medicine Image (Optional)" with a small camera icon and the text "Upload image from device". Below this are two inputs: "Price \*" with the value "60" and "Available Quantity \*" with the value "1". At the bottom are two buttons: a grey "Back" button and a green "Save Medicine to Inventory" button.

Add Medicine (Step 1)

Admins add medicines to the global catalogue with name and category.



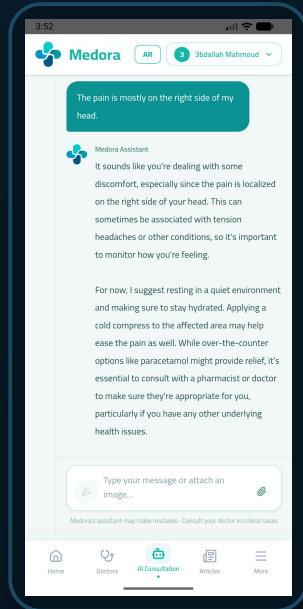
The screenshot shows the same "Add New Medicine" modal as in Step 1, but with more data entered. The "Medicine Name \*" field now contains "Coronol". The "Category" dropdown is now set to "Cold & Flu" with a small downward arrow. The "Active Ingredient" field now contains "Pseudoephedrine HCl". In the "Your Inventory Details" section, the "Medicine Image (Optional)" field now shows a small red and white box icon with the text "Image uploaded" and a small trash icon to its right. The "Price \*" field still shows "60" and the "Available Quantity \*" field still shows "1". The "Back" and "Save Medicine to Inventory" buttons remain at the bottom.

Add Medicine (Step 2)

Inventory details, image upload and available quantity are confirmed.

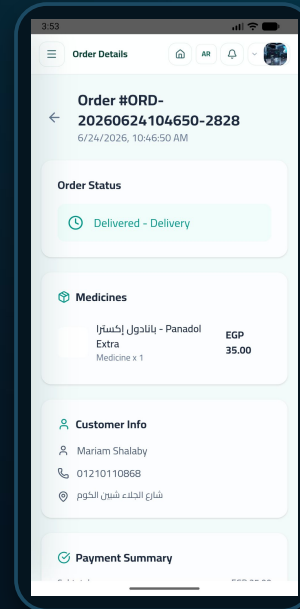


# Native Mobile App : Patient



**AI Medical Assistant**

Safe, bilingual medical chat on mobile

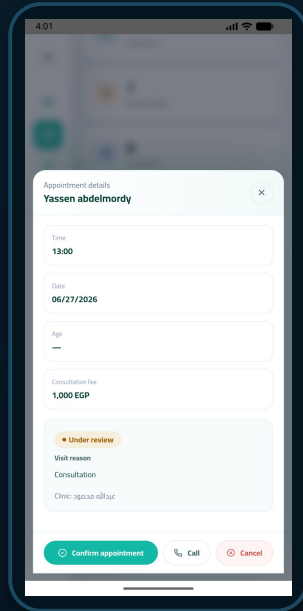


**Order Tracking**

Live delivery status & order details

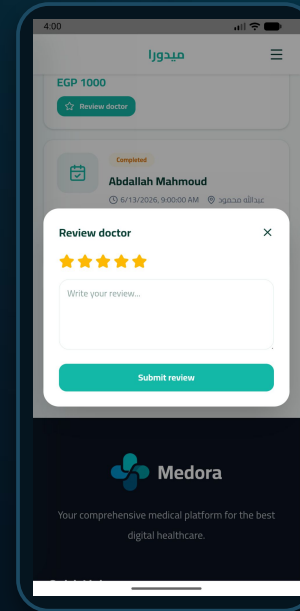


# Native Mobile App : Booking & Reviews



## Appointment Details

Confirm or cancel a booked visit

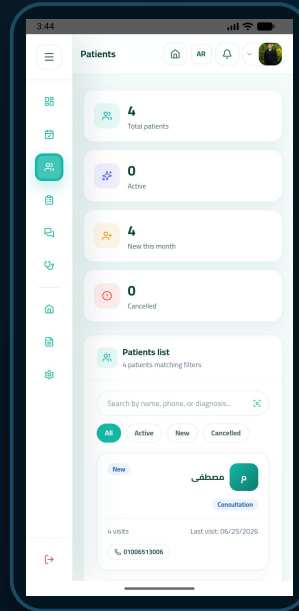


## Reviews

Rate a doctor in the full Arabic interface



# Native Mobile App : Provider



## Doctor Dashboard

Patient list, stats and quick filters on mobile



# Where Medora Goes Next



## Online Payments

Card and mobile-wallet options (Fawry, Vodafone Cash) beyond cash on delivery.



## Telemedicine

Embedded video consultations for remote and mobility-limited patients.



## Real-Time Notifications

Push delivery via WebSockets / SignalR, replacing periodic polling.



## Smarter AI Retrieval

Vector-based semantic search with embeddings to improve medical grounding.



## Delivery Rider App

A courier app with live GPS so patients can track their delivery on a map.



## Lab Test Ordering

Order lab tests from certified labs with digital results to patient and doctor.



# Business Model & Market

## Revenue Streams



### Booking Commission

A small fee per completed appointment booked through the platform.



### Pharmacy Order Fee

A percentage on each fulfilled medicine order routed to a pharmacy.



### Provider Subscriptions

Monthly plans giving clinics & pharmacies dashboards, analytics and priority listing.



### Featured Placement

Optional paid promotion for verified providers in search results.

## Market Opportunity

Egypt's fragmented outpatient and pharmacy market is largely offline. Digital coordination is already proven in banking and food delivery, healthcare is the next vertical to be digitised.



**Patients** Free to use, the demand side and growth engine.



**Clinics & Doctors** Independent practices needing simple booking tools.



**Pharmacies** Local pharmacies gaining a new digital order channel.



# Go-to-Market Strategy



## Phase 1 : Local Beachhead

Launch in one city. Onboard a focused cluster of verified clinics and nearby pharmacies so patients always find availability and stock.



## Phase 2 : Demand Growth

Drive patient adoption with the free app, the safe AI assistant and digital prescriptions as the hook. Word of mouth + targeted social campaigns.



## Phase 3 : Provider Monetisation

Convert active clinics and pharmacies to paid subscriptions and featured placement once they see real booking and order volume.



## Phase 4 : Geographic Expansion

Replicate the city playbook across governorates, then extend scope (payments, telemedicine, lab tests) from the existing schema.



## CONCLUSION

# What Medora Demonstrates



**A Real, Complete System** A production-quality platform that solves five documented outpatient problems end-to-end.



**End-to-End Coordination** Discovery, booking, prescription, ordering and delivery operate as one linked flow.



**Safety as a Primary Requirement** The medical AI treats safety constraints as core engineering, not a marketing feature.



**Bilingual From the Foundation** Arabic and English are equal first-class languages with full right-to-left support.

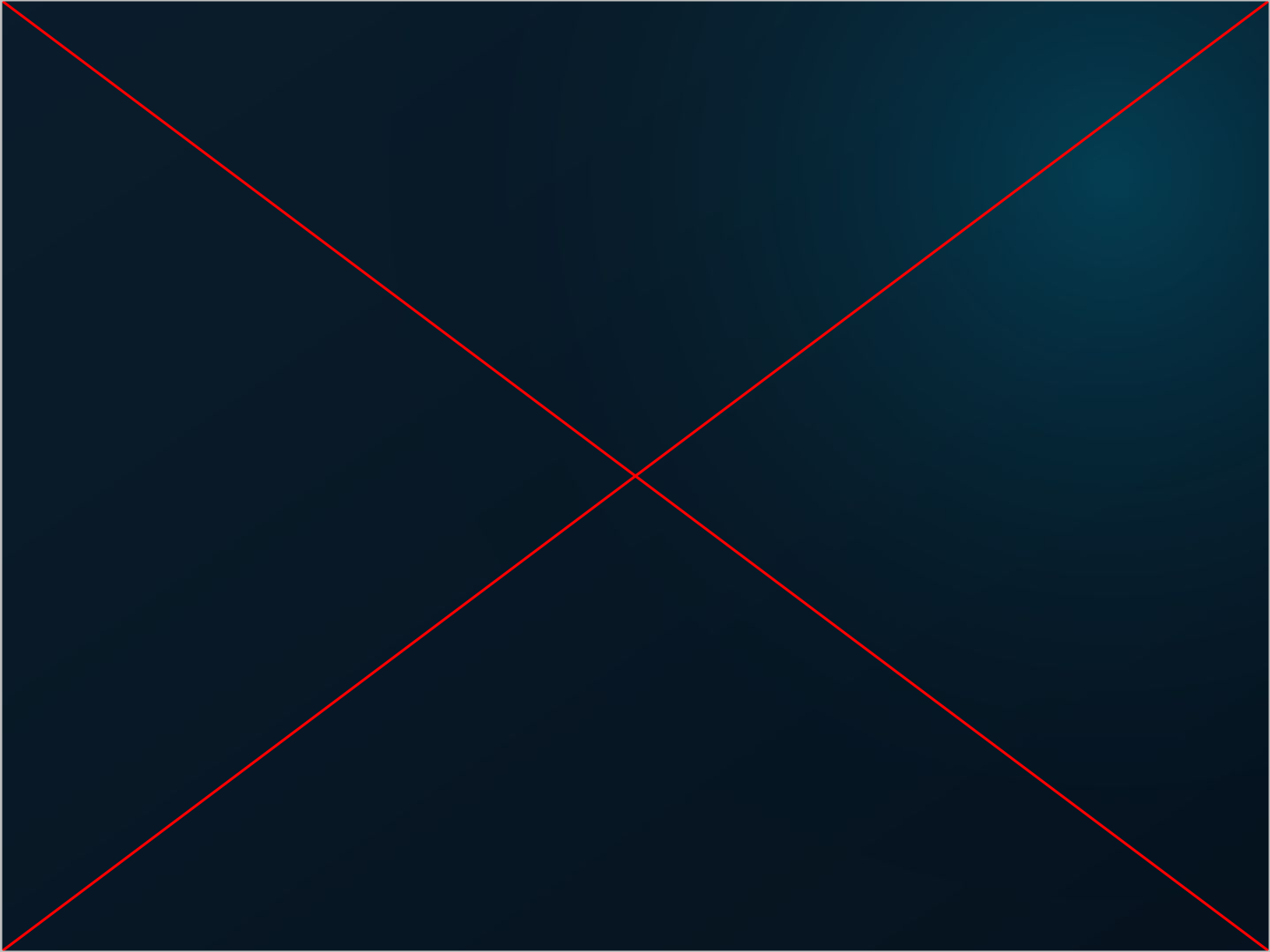


**Built on Safety & Trust** Every feature was designed with patient safety and provider trust as constraints.





**Live Demo**





**Thank You**